

1

Bottlenecks

- Autoregressive drafters model dependencies well but their latency grows with proposal length.
- Parallel drafters propose long blocks in one pass, yet independent positions create multimodal collisions and rapid suffix acceptance decay.
- Fixed-length verification wastes scarce target batch capacity on risky suffix tokens under high concurrency.

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Scheduler

MAXIMIZE LIVE THROUGHPUT

- Profile engine steps/second as a function of verification batch size once at initialization.
- Globally sort prefix extensions by survival probability and greedily admit them while expected throughput increases.
- Early stopping preserves non-anticipation and exact target distribution; asynchronous production scheduling hides its latency.

2

Semi-autoregressive

- A DFlash-style parallel backbone produces all hidden states and base logits in one heavy forward pass.
- A lightweight sequential head adds prefix-dependent transition biases while preserving exact per-token softmax probabilities.
- Markov and RNN variants inject one-step or full-block history; the default low-rank Markov head uses rank 256.

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Evidence

- On Qwen3 4B/8B/14B, accepted length improves over Eagle3 by 30.9/26.7/30.0% and over DFlash by 16.3/18.4/18.3%.
- The sequential head adds only 0.2–1.3% round latency while delivering up to 30% more accepted tokens.
- DeepSeek-V4 live traffic gains 60–85% per-user speed on Flash and 57–78% on Pro at matched practical throughput.

3

Confidence

PREDICT PREFIX SURVIVAL

- A small head predicts conditional acceptance c_k from the backbone state and previous-token embedding.
- $c*_k = 1 - 0.5 \ ||p_{d,k} - p_{t,k}||_1$; $a_{r,j} = \text{product}_{(i \leq j)} c_{r,i}$
- Sequential Temperature Scaling calibrates cumulative survival probability, reducing average ECE from 3–8% to about 1%.

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Limits & release

- The scheduler expands verification under light load and prunes aggressively when target capacity saturates.
- The parallel backbone still pays a fixed cost to draft the full block; low-acceptance hard queries may never recover it.
- Future difficulty-aware early exit could bypass full drafting; checkpoints and the DeepSpec training repository are released.

DSPARK

DSpark

Confidence-Scheduled Speculative Decoding with Semi-Autoregressive Generation | 2026

Draft long blocks with a powerful parallel backbone, restore local token dependencies with a tiny sequential head, then spend target-model verification only where calibrated survival probability justifies the current system cost.

speculative decoding

semi-autoregression

serving